Review Pulse

How we trained two sentiment classifiers on 8,000 Amazon reviews and built a live web app to compare them.

INTELLIGENT SYSTEMS (ISY503)
ASSESSMENT 3

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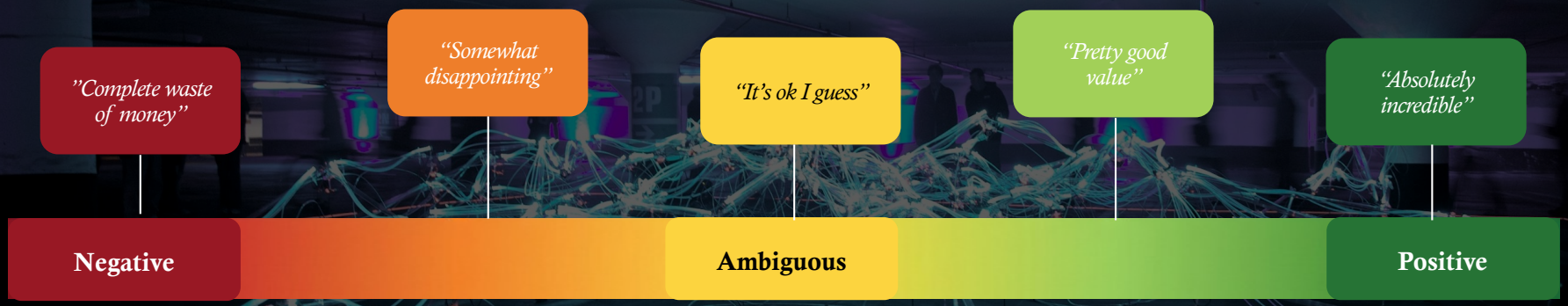
PROBLEM STATEMENT

The Sentiment Spectrum – what our model must learn to classify



- Sentiment analysis: classifying text as positive or negative opinion
- Real-world value: product feedback loops, brand monitoring, recommendation systems
- Challenge: reviews vary in **length** (2–800 words), **domain**, **writing style**, **negation**, and **sarcasm**
- Goal: build, compare, and deploy two approaches — classical ML vs neural

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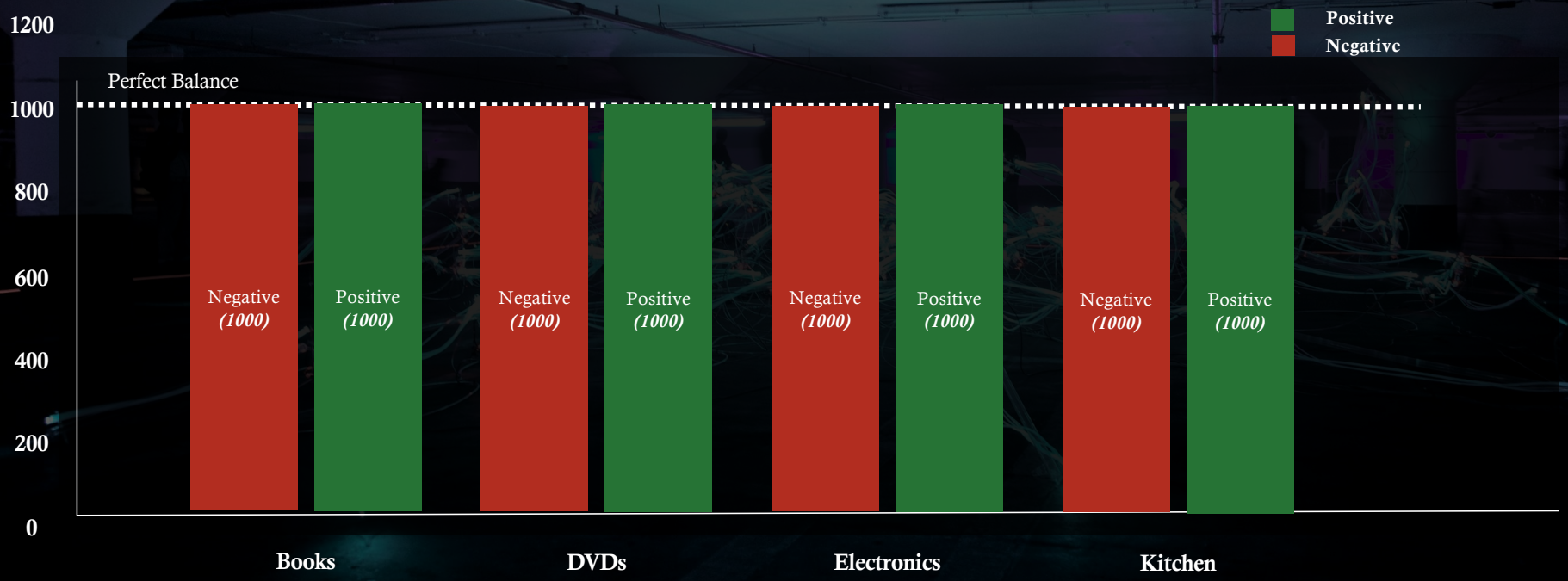
DATASET

Balance – 1,000 reviews per domain per class



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PREPROCESSING PIPELINE

Raw .review Files – pseudo-XML format

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BeautifulSoup Parser – extract review_text + rating

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Label Audit – flag 3-star and rating/label conflicts

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clean_text – lowercase, HTML strip, negation expansion, remove punctuation

Wasn't good → was not good

↓

remove_outliers – drop < 10 or > 500 words (279 removed)

↓

Stratified Split – 70% train / 15% val / 15% test (seed = 42)

↓

train = 5,404 / val = 1,158 / test = 1,159

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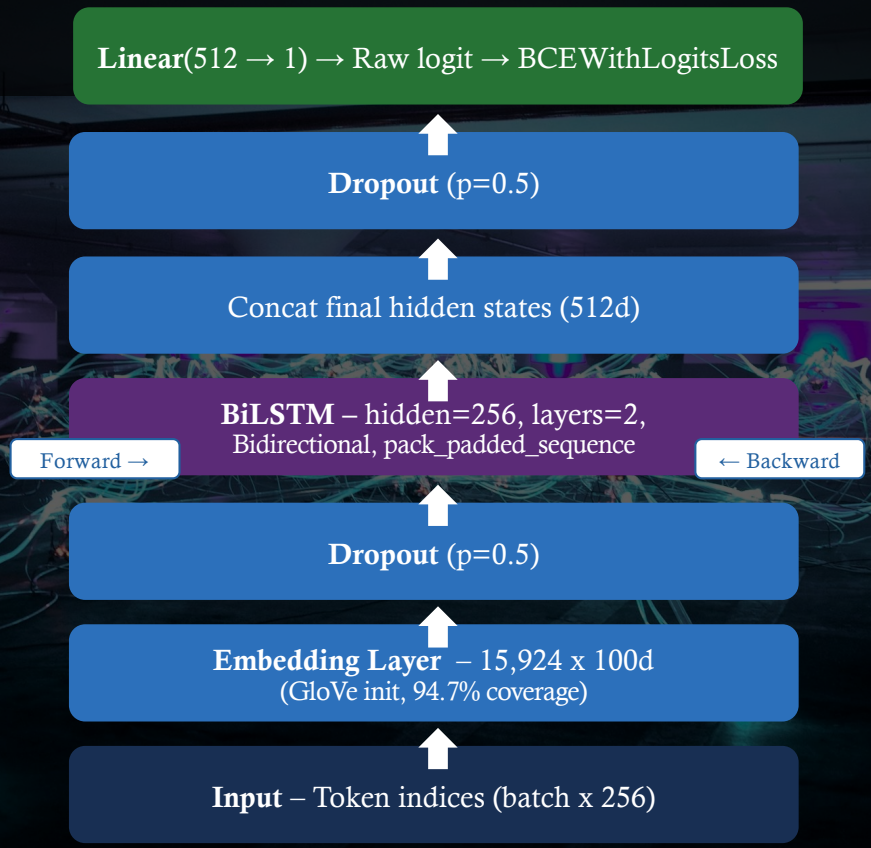
ARCHITECTURE OVERVIEW

BiLSTM + GloVe



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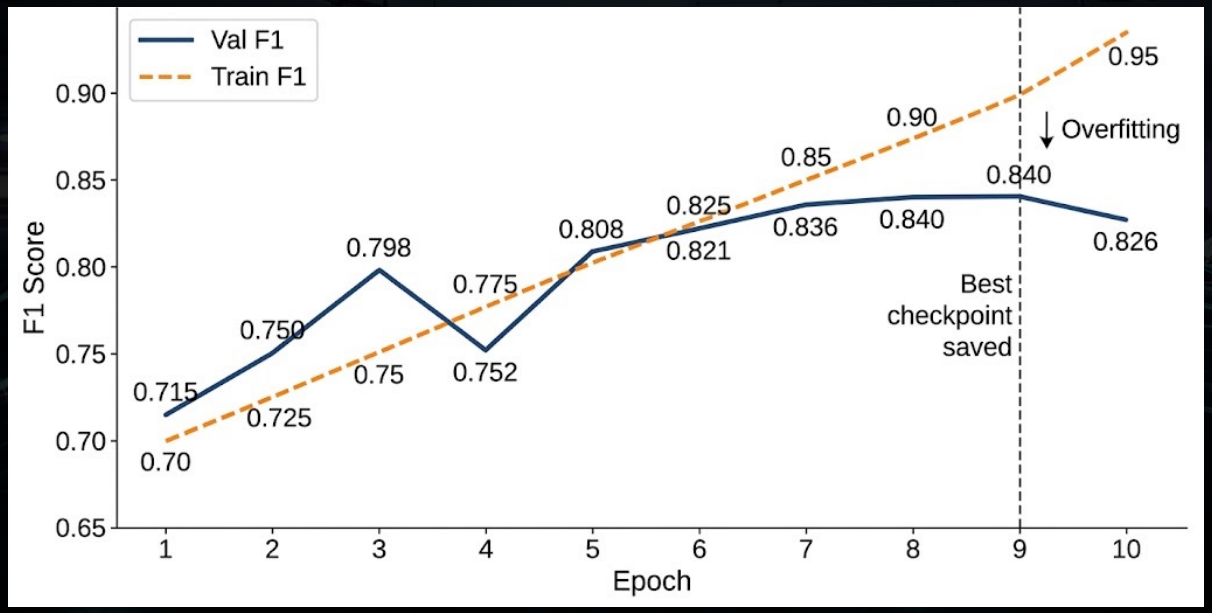
TRAINING

BiLSTM Training – Val F1 over 10 Epochs



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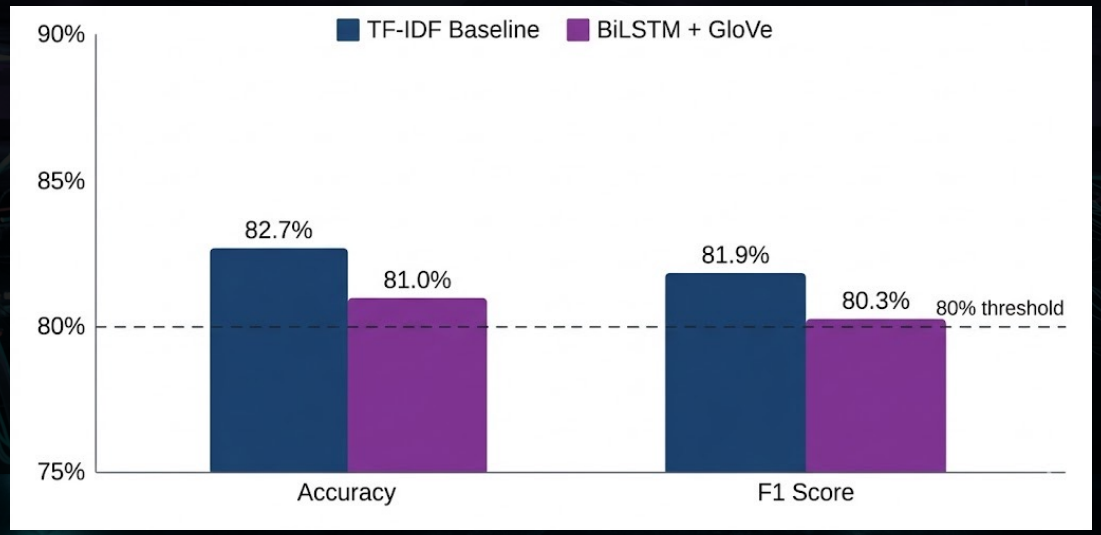
RESULTS

Model Comparison



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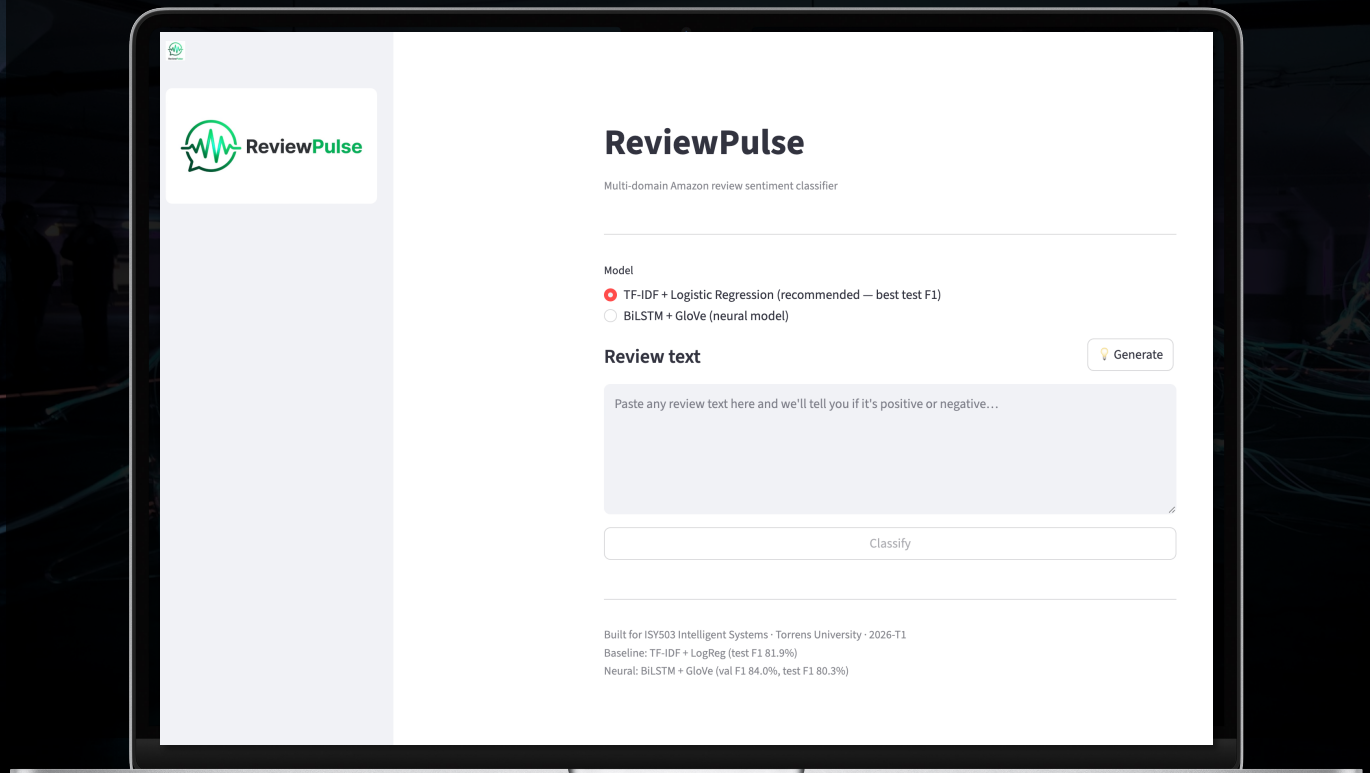
Error Analysis

Failure Type	*Example Review*	*Root Cause*
Negation	"Not bad at all"	Double negation confuses bag-of-words and sequence models
Sarcasm	"Oh, great, stopped working after a week"	Irony not represented in training labels
Out-of-distribution	Delivery logistics review	Domain shift beyond the 4 trained categories

LIVE DEMO



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Ethics & Limitations

Concern	*Detail*
Label quality	Labels from filenames, not human raters - audited explicitly: 0 ambiguous rows
Domain bias	4 product categories only - generalisation to other domains is untested
Negation & Sarcasm	Systematic failure mode in both models
Confidence calibration	BiLSTM logits are uncalibrated - 99% ≠ 99% accuracy
Dataset age	Blitzer et al. 2007 - ~20-year-old reviews; language patterns may have shifted
Deployment risk	App should not be used for high-stakes decisions without human review

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Future Work

Next steps:

- DistilBERT / RoBERTa — contextual embeddings for negation and sarcasm
- Confidence calibration — Platt scaling or temperature scaling on BiLSTM logits
- More domains — extend beyond the 4 Blitzer categories
- Explainability — LIME or attention visualisation to show why the model predicted each label

SUMMARY / QUESTIONS

Key Outcomes Unlocked:

- Full pipeline: raw data → preprocessing → two trained models → live web app
- Classical model wins on test — TF-IDF F1 81.9% vs BiLSTM 80.3%
- Neural model demonstrates architecture — satisfies rubric, shows ML maturity
- 117 unit tests | real error analysis | ethics documented
- GitHub: github.com/lfariabr/review-pulse

Thank you!

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